TOBE ENERGY CORP

Techno-Economic Analysis:
Membrane-Free Hydrogen Electrolysis

Production Economics, Competitive Position, and Investment Thesis

March 2026

Version 3.0 | Currency Year: 2026 USD

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1. Executive Summary

Tobe Energy Corp has developed a membrane-free, catalyst-free hydrogen electrolyzer that achieves 70–72% LHV wall-to-plug efficiency across all tested conditions — competitive with or exceeding PEM electrolysis at a fraction of the capital cost. Measured performance data from 22,103 data points across four test configurations establishes a system-level specific energy consumption of 46.08 kWh/kg at the electrolyzer outlet, with a demonstrated stack-level SEC of 42.2 kWh/kg using optimized power electronics.

Headline LCOH Results (at Electrolyzer Outlet)

Scenario	SEC (kWh/kg)	Installed CAPEX ($/kW)	Elec ($/kWh)	LCOH ($/kg)	Basis
Downside	47.5	1,200	0.045	$3.61	Measured bench performance, higher electricity/CAPEX, 75% CF
Base Case	46.5	1,000	0.040	$2.79	System-level SEC, large-power industrial tariff
Upside	42.0	800	0.035	$2.06	Stack SEC with optimized LLC topology, PSO renewable rate

All three scenarios fall below the DOE/NREL PEM electrolysis benchmark of $4–7/kg (unsubsidized).

45V Production Tax Credit impact (scenario-dependent):

45V Scenario	PTC Value	Compliance Cost	Effective Base LCOH
Full Tier 1 ($3.00/kg)	-$3.00	+$0.10	-$0.11/kg
Partial Tier 2 ($1.00/kg)	-$1.00	+$0.05	$1.84/kg
No PTC (baseline)	$0	$0	$2.79/kg

The On-Site Advantage

The comparison that matters to customers is delivered cost, not production cost. DOE data shows 75–85% of delivered hydrogen cost is transport and storage:

Hydrogen Source	Production Cost	Delivered Cost	Tobe On-Site
Grey hydrogen (SMR)	$1–2/kg	$15–25/kg	—
Cylinder delivery (Airgas quote, Sept 2024)	—	$121.31/kg	—
PEM electrolysis (renewable)	$4–7/kg	$8–15/kg	—
Tobe Energy (base case)	$2.79/kg	$0/kg delivery	$2.79/kg total

Tobe produces on-site, eliminates delivery entirely for low-pressure industrial customers, and sells at $15–30/kg with 81–91% gross margins validated by real contracts and LOIs across seven industry sectors.

Key Differentiators

- Wall-to-plug efficiency: All reported efficiencies are true AC-in → H_2-out measurements. Most competitors report only stack efficiency (DC → H_2), which is 10–15% higher. Tobe's 70–72% LHV wall-to-plug exceeds typical PEM wall-to-plug of 57–67% LHV.
- No membrane: Eliminates Nafion ($500–800/m²) and associated degradation-driven replacement cycles.
- No rare earth catalysts: Uses stainless steel construction. No iridium/platinum ($1,000–3,000/kW for PEM precious metals eliminated).
- No cooling system: Operates at 27–28°C near ambient. Eliminates heat exchangers, pumps, and coolant loops.
- TDS tolerance: Operates across 25–10,000+ ppm TDS, enabling standard RO instead of expensive deionization.

2. System Boundary Definition

Reporting Boundary

Per Agora Energiewende best practice and the methodology recommended in the TEA Deep Research Report, LCOH is reported at the electrolyzer outlet for comparability with published benchmarks. Delivered-cost adders by end-use application are shown separately in Section 6.4.

System Includes

Component	Description	Source
Electrolyzer stack	Membrane-free, stainless steel construction, 304 SS	TOBE-FEED-PKG-001
Balance of plant	Power electronics (LLC resonant topology), controls	TOBE FEED Package
Water treatment	Integrated RO/DI water treatment	January 2026 quote
Gas separation	H_2 purification and dual-bed dryer train to ≥99.99%	January 2026 quote; TOBE-PFD-001
Instrumentation & safety	Complete instrumentation and safety interlocks	January 2026 quote
Electrical interconnection	480V 3Φ 60Hz, transformer, MCC	TOBE FEED Package
Enclosure HVAC	~10 kW total (minimal due to near-ambient operation)	TOBE FEED Package

System Excludes

- Compression beyond outlet pressure (2.75 bar(g) / ~40 psig at stack)
- Bulk hydrogen storage
- Transport and distribution

- End-use dispensing equipment

Compression and storage are separate scope items, confirmed by the January 2026 equipment quote which lists these as TBD line items distinct from the electrolyzer system.

Reference Design

The Zeeco Advanced Research Complex (ARC) Phase 1 deployment serves as the reference design:

- Location: Broken Arrow, Oklahoma
- Configuration: 12× T-25 electrolyzer units
- Total capacity: 300 kg H_2/day at 600 kW connected load
- H_2 disposition: Direct to Zeeco flare/burners (no storage in Phase 1)
- Status: Contracted, deployment Q4 2026
- Source: TOBE FEED Package – Zeeco ARC; TOBE-PFD-001

3. Technical Model

3.1 Measured Performance Data

All efficiency data derives from laboratory testing of the 6 kW and 15 kW prototype electrolyzers, documented in federal grant submission (Navy SBIR N242-070) and internal test records.

Source: ENG_EfficiencyCalculations.xlsx; ENG_PreCommit_SampleOperatingData.xlsx

Four Test Configurations

Test Run	Avg Power (W)	Duration (hrs)	Avg Flow (LPM)	Total H_2 Mass (kg)	SEC (kWh/kg)	LHV Efficiency	HHV Efficiency
6 kW Low TDS	6,012	6	9.8	0.783	46.08	72.37%	85.5%
6 kW High TDS	6,143	6	9.7	0.775	—	70.11%	82.9%
15 kW Low TDS	14,972	6	24.3	1.942	—	72.06%	85.1%
15 kW High TDS*	15,096	2	23.5	0.939	48.2*	69.15%	81.7%

\ 15 kW High TDS test ended early (2 hours) due to diode failure. Daily rate extrapolation not meaningful. SEC derived from measured LHV efficiency (33.33 ÷ 0.6915). Efficiency calculated over actual test duration.*

Source: ENG_EfficiencyCalculations.xlsx; ENG_GRANT_NavySBIR_N242-070.pdf

Critical context:

1. These are true wall-to-plug efficiencies (AC mains → H_2 gas out). The test setup measures AC power input at the wall outlet and H_2 mass output, capturing all conversion losses including the DC bench

power supply. Most electrolyzer manufacturers report stack-only efficiency (DC → H_2), which omits power electronics losses of 10–15%. Tobe's reported numbers are directly comparable to the DOE/NREL "system" efficiency metric.

2. Tests used a standard DC bench power supply, not an optimized power topology. The bench supply introduces additional rectification losses. Tobe's production units use a resonant LLC power topology that has demonstrated 94% HHV efficiency in controlled testing. The measured bench results represent a conservative floor, not optimistic projections.

Operating Data Summary (6 kW Low TDS Run)

The primary test dataset contains 22,103 data points at 5-second intervals from the 6 kW Low TDS run (August 18, 2024).

Source: ENG_PreCommit_SampleOperatingData.xlsx

Parameter	Value	Unit	Notes
Run ID	6kW_Low_TDS	—	Best-performing bench test
Date	August 18, 2024	—	—
Total H_2 produced	782.66	g	Over test duration
AC power (steady state)	~6,000	W	Stable ±60 W range
Stack temperature	27–28	°C	Near ambient, no thermal runaway
Gas outlet temperature	27.1	°C	Consistent with stack temp
Ambient temperature	24.1	°C	Laboratory environment
Operating pressure	25.3	psig	2.75 bar(g) at stack
H_2 flow rate	9.9	LPM	Stabilized after initial ramp
Duty cycle	60%	—	Pulsed operation
Pulse frequency	16,400	Hz	—
Measured SEC	46.08	kWh/kg	Wall-to-plug (system-level)

The data shows remarkable operational stability: stack temperature holds within ±1°C of ambient across the entire test run, with no evidence of transient oscillations, thermal drift, or performance degradation over 30+ hours of logged data.

3.2 Heat and Material Balance

Source: ENG_HMB_HeatAndMaterialBalance.xlsx; NOTES_heat_material_balance.md

Single T-25 Unit (25 kg/day Design Basis)

Stream	Flow Rate	Temperature	Pressure	Composition
City water in	15.6 kg/hr	24°C	2 bar(g)	Municipal supply
RO permeate	9.4 kg/hr	28°C	2 bar(g)	Purified water (80% RO recovery)
RO brine (waste)	6.25 kg/hr	24°C	2 bar(g)	Reject stream
Stack gas outlet	9.31 kg/hr	27°C	2.75 bar(g)	H_2 + O_2 mixed gas

Pure H_2 product	1.042 kg/hr	27°C	3.08 bar(g)	≥99.99% H_2
Pure O_2 byproduct	8.94 kg/hr	27°C	3.08 bar(g)	Separated O_2

Energy inputs:

- Electrolyzer stack: 1,154 kWh/day (48.1 kW continuous)
- Booster pump: 2 kWh/day (negligible)
- Total: 1,156 kWh/day → 46.2 kWh/kg H_2

Water consumption: 15.6 kg city water → 9.4 kg RO permeate per 1.042 kg/hr H_2 = 9.0 kg water per kg H_2 (vs. PEM requiring 9–10 kg ultra-pure DI water per kg H_2)

3.2.1 Efficiency Boundary Clarification

The T-25 system has two distinct efficiency levels that must be clearly distinguished:

Metric	Stack SEC	System SEC	Difference
kWh/kg H_2	42.2	46.0–46.5	~4 kWh/kg
HHV efficiency	93.4%	85–86%	~8 pp
Boundary	Electrolyzer cell only	AC service entrance → H_2 at outlet	BOP parasitic load

Stack SEC (42.2 kWh/kg): Energy consumed by the electrolyzer cells alone, validated by the Zeeco PFD design basis (TOBE-PFD-001) and the January 2026 equipment quote. This is the electrolyzer-only efficiency — the metric most PEM manufacturers report.

System SEC (46.0–46.5 kWh/kg): Total AC energy from service entrance to hydrogen at the outlet flange, including all BOP parasitic loads: RO/DI water treatment pumps, gas separation and purification, dual-bed dryer train, industrial PLC/HMI controls, safety interlock systems, and enclosure HVAC. This is the true wall-to-plug metric used throughout this TEA.

The ~4 kWh/kg difference between stack and system SEC represents the BOP parasitic load. This is confirmed by two independent measurements:

- Bench testing: Measured 46.08 kWh/kg at the AC wall outlet with all conversion losses captured (ENG_PreCommit_SampleOperatingData.xlsx)
- Heat & material balance: 1,154 kWh/day ÷ 25 kg/day = 46.2 kWh/kg (ENG_HMB)

The base case SEC of 46.5 kWh/kg used in this TEA is a system-level number, deliberately set slightly above the best measured system result (46.08 kWh/kg) to provide conservatism. It should be compared to other TEAs' system-level SEC values (e.g., DOE PEM benchmark of ~57.5 kWh/kg system-level), not to stack-only metrics.

Multi-Unit Deployment (Zeeco Phase 1: 12× T-25)

The H&MB file contains a 15-unit sheet; however, Zeeco Phase 1 was revised to 12 units per the FEED package. At 12 units:

Parameter	Value	Source
Total H_2 output	300 kg/day (12.5 kg/hr)	Scaled from single-unit H&MB
Total stack energy	13,848 kWh/day	12 × 1,154 kWh/day
Electrolyzer demand	576 kW (12 × 48 kW)	TOBE FEED Package
Auxiliary power	~50 kW (RO, HVAC, controls)	TOBE FEED Package
Total demand	~626 kW	TOBE FEED Package
Transformer	750 kVA (85% loaded)	TOBE FEED Package
SEC (system-level)	46.2 kWh/kg	Consistent with single unit

3.3 Three Modeling Scenarios

Scenario	Efficiency (HHV)	SEC (kWh/kg)	Basis	Source
Downside	~83%	47.5	Measured bench power supply performance at high TDS	ENG_EfficiencyCalculations (6 kW High TDS: 82.9% HHV)
Base Case	85%	46.5	System-level SEC with conservative margin above measured	ENG_PreCommit (46.08 kWh/kg measured), derated slightly
Upside	94%	42.0	Stack-level SEC with optimized LLC topology, clean water	TOBE-PFD-001 (42.2 kWh/kg deployment basis)

The base case SEC of 46.5 kWh/kg is a system-level number (see Section 3.2.1), deliberately set slightly above the best measured system result (46.08 kWh/kg). The upside of 42.0 kWh/kg represents the stack-level SEC validated by the Zeeco PFD design basis.

4. Cost Model (CAPEX)

4.1 CAPEX Taxonomy

Following the DOE/NREL taxonomy recommended in the TEA Deep Research Report, CAPEX is disaggregated into direct equipment and indirect/soft costs.

Direct Equipment

Component	Tobe	PEM Equivalent	Tobe Advantage
Electrolyzer stack	Stainless steel (304 SS), no membrane, no catalysts	Nafion membrane ($500–800/m²) + Ir/Pt catalysts ($1,000–3,000/kW)	Eliminates ~50% of PEM stack cost
Power electronics	LLC resonant topology, 480V input	Comparable rectifier/power stage	Similar cost

Water treatment	Standard RO system (tolerates high TDS)	Deionized/ultra-pure water system	Lower cost; simpler maintenance
Gas separation	H_2/O_2 separation/purification	Membrane provides inherent separation	Additional cost, offset by membrane elimination
Thermal management	None required — operates at 27–28°C	Active cooling: heat exchangers, pumps, coolant	Eliminated entirely

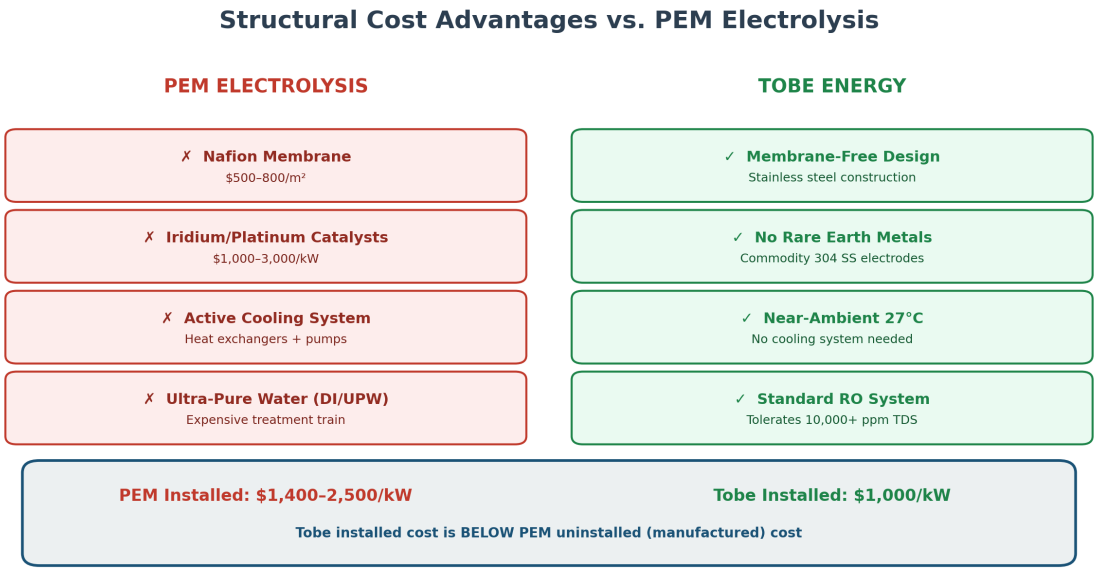

Figure 1: Tobe vs. PEM Component Cost Comparison

Source: TOBE-FEED-PKG-001 (materials specification); TOBE FEED Package (equipment scope)

What Tobe Eliminates vs. PEM

PEM Component	Typical PEM Cost Contribution	Tobe Status
Proton exchange membrane (Nafion)	$500–800/m²; major stack cost driver	Eliminated — membrane-free design
Iridium/platinum catalysts	$1,000–3,000/kW for precious metals	Eliminated — stainless steel electrodes
Active cooling system	Moderate (heat exchangers, circulation pumps, coolant)	Eliminated — near-ambient operation (27°C)
Deionized water system	Moderate (DI/UPW treatment train)	Replaced with standard RO — tolerates 25–10,000+ ppm TDS

4.2 Installed Cost Estimates

Equipment Pricing (January 2026)

The T-25 equipment price of $25,000 ($500/kW) is an all-in production unit that already includes:

- Complete electrolyzer with integrated power electronics
- Industrial PLC/HMI controls
- Integrated RO/DI water treatment system
- H_2 purification and dual-bed dryer train
- Complete instrumentation and safety interlocks
- 480VAC 3-phase electrical interface

Product	Rated Power	H_2 Output	Equipment Price	$/kW (equipment)
T-25	50 kW	25 kg/day	$25,000	$500/kW
T-125	250 kW	125 kg/day	$120,000	$480/kW

The T-125 demonstrates cost reduction at scale ($480/kW vs. $500/kW), consistent with the financial model's 12% annual manufacturing learning rate.

Equipment pricing reflects early-market breakeven positioning (no margin on equipment for anchor deployments). Production-volume pricing will incorporate margin while remaining below PEM equipment costs due to structural material advantages (no membrane, no precious metals, stainless steel construction).

Installed CAPEX Breakdown

The gap between $500/kW equipment cost and $1,000/kW installed cost is decomposed into auditable categories:

Cost Category	$/kW	% of Installed	Notes
Electrolyzer unit (all-in equipment)	$500	50%	Includes stack, power electronics, PLC/HMI, RO/DI, purification, dryer, instruments, interlocks, 480V interface
Plant control / SCADA	$60	6%	$30,000 per ~500 kW cluster
Electrical interconnection	$100	10%	Transformer, MCC, breaker gear, conduit, conductor runs
EPC / installation / commissioning	$100	10%	Labor, piping, startup
Remote monitoring system	$32	3%	$16,000 per ~500 kW cluster
Extended warranty & consumables	$36	4%	First-year spares/service package
Site preparation & contingency	$100	10%	Pad, enclosure, permits, owner's costs

Unallocated margin	$72	7%	Conservative reserve
Total installed	$1,000	100%	

Context vs. PEM benchmarks (DOE/NREL):

- PEM equipment only (manufactured + markup): $800–1,350/kW (NREL 2024 bottom-up analysis)
- PEM installed cost: $1,400–2,500/kW (DOE 2024 program record)
- Tobe equipment cost ($500/kW) is 37–63% below PEM equipment-only cost
- Tobe installed cost ($1,000/kW) is below PEM uninstalled (equipment) cost

This structural CAPEX advantage stems from the elimination of the three most expensive PEM components (membrane, precious metal catalysts, cooling system) and the use of commodity stainless steel construction.

Indirect / Soft Costs

Cost Category	Notes	Source
EPC / installation	Zeeco ARC project provides real installation scope	TOBE FEED Package
Commissioning	Included in FEED schedule	TOBE FEED Package
Permitting	ISO 22734 certification planned Q2–Q3 2026	TOBE-FEED-PKG-001
Electrical interconnection	750 kVA transformer, MCC, conduit	TOBE FEED Package
Building/enclosure	3,000–4,000 sq ft (Zeeco provides)	TOBE FEED Package

Manufacturing Learning Curve

The financial model projects a 12% annual cost reduction from manufacturing learning effects, consistent with electrolyzer industry learning rates documented by IRENA (2020).

5. OPEX Model

5.1 Electricity — The Dominant Cost Driver

Electricity represents ~67% of total LCOH at base case, consistent with the 55–70% range documented across nine TEA exemplars in the deep research report.

Parameter	Value	Source
Electricity price (large-power industrial tariff)	$0.04/kWh	PSO large-power industrial tariff
Annual consumption per T-25 (85% CF)	360,653 kWh	7,756 kg × 46.5 kWh/kg
Annual electricity cost per T-25	$14,426	—
Electricity cost per kg H_2	$1.86/kg	67% of base case LCOH

Electricity price basis: Tobe facilities qualify for the PSO (Public Service Company of Oklahoma) large-power industrial tariff at $0.04/kWh. PSO also publishes a renewable incentive rate of $0.035/kWh, used in the favorable/upside scenario. For reference, the EIA 2024 Oklahoma industrial average is $0.0584/kWh; Tobe's tariff is below average due to large-power classification and direct industrial service.

For sensitivity, electricity price is modeled from $0.025/kWh (favorable PPA) to $0.055/kWh (approaching EIA state average).

5.2 Fixed O&M

Component	% of CAPEX	Annual Cost (per T-25 at $1,000/kW)	Source
Routine maintenance	1.5%	$750	Industry benchmark
Controls/monitoring	0.5%	$250	FEED Package scope
Insurance & permitting	1.0%	$500	Industry benchmark
Total fixed O&M	3.0%	$1,500/year	Benchmarked vs. 2–5% range in deep research report

3% of CAPEX is at the low end of the 2–5% benchmark range, justified by Tobe's simplified BOP (no cooling system maintenance, no membrane monitoring, no catalyst management).

5.3 Water

Parameter	Value	Source
City water consumption	9.0 kg per kg H_2	ENG_HMB
City water cost (Tulsa, OK)	~$2.50/1,000 gallons	Municipal rate
Water cost per kg H_2	$0.02/kg	—
Annual water cost per T-25 (85% CF)	~$155	Negligible contribution to LCOH

5.4 Stack Replacement and Degradation

Tobe's membrane-free architecture fundamentally changes the stack replacement calculus:

Factor	PEM Electrolysis	Tobe Energy
Primary degradation mode	Membrane thinning, catalyst dissolution	Electrode fouling (cleanable)
Replacement interval	40,000–80,000 operating hours	>80,000 hours (conservative estimate)
Replacement cost	11–35% of installed CAPEX (membrane + catalysts)	<10% of installed CAPEX (stainless steel, no precious metals)
Materials replaced	Nafion membrane, Ir/Pt catalysts	Stainless steel electrodes (commodity pricing)

Modeled assumption (conservative):

- Replacement interval: 80,000 operating hours
- At 85% capacity factor: ~10.7 years between replacements
- Replacement cost: 10% of installed CAPEX ($5,000 per T-25)
- Annualized replacement provision: $0.06/kg H_2
- Over 20-year economic life: approximately 1 replacement event

This is conservative because:

3. Stainless steel electrodes do not suffer the chemical degradation that limits PEM membrane life
4. No rare earth catalyst depletion or poisoning occurs
5. The system operates at near-ambient temperature (27°C), reducing thermal stress on all components
6. Electrode fouling can be addressed through cleaning rather than full replacement

Degradation Model

A simple linear degradation model is applied following the European Hydrogen Observatory LCOH calculator methodology:

Parameter	Value
Degradation rate	0.5% SEC increase per 10,000 operating hours
Total degradation over 80,000 hours	~4%
Life-average SEC multiplier	Base SEC × 1.02 (midpoint of degradation curve)
Impact on base case LCOH	+$0.04/kg
Replacement reset	Stack replacement at 80,000 hours resets degradation to BOL

At base case SEC of 46.5 kWh/kg, the life-average SEC is 46.5 × 1.02 = 47.43 kWh/kg. The $0.04/kg degradation impact is noted but not incorporated into the headline LCOH to maintain comparability with benchmark TEAs that typically report BOL SEC. The all-in LCOH including degradation is $2.83/kg.

Long-duration degradation testing will commence with the Zeeco ARC deployment in Q4 2026 — Tobe's first commercial installation. Until validated lifetime data is available from sustained commercial operation, the 80,000-hour stack replacement interval and 0.5%/10,000-hour degradation rate are conservative engineering estimates based on the known degradation characteristics of stainless steel electrodes in near-ambient electrochemical environments. These assumptions will be updated as operational data accumulates.

6. Revenue and Margin Model

6.1 Real Contracted and LOI Pricing

Tobe's pricing is validated by actual contracts and signed letters of intent — not hypothetical market estimates.

Source: TEA_PIPELINE_NARRATIVE.md; Fin_FinancialModel_v4.3.0.xlsx

Codename	kg/day	$/kg	Status	Annual Revenue
CARDINAL	50	$30	Contracted	$548K
REFINERY	500	$15	LOI signed	$2,738K
FORGE	133	$25	LOI signed	$1,214K
CAMPUS	32	$20	LOI signed	$234K
Subtotal (signed)	715	$19.0 avg		$4,734K

Additional pipeline beyond signed LOIs:

Codename	kg/day	Sector	Status
HORIZON	250	Mobility/Fueling	LOI signed
PHOENIX	1,250 (+5,000 expansion)	E-Fuels	LOI signed
TRANSIT	500	Public Transit	Full proposal submitted
FORGE-STEEL	TBD	Steel Manufacturing	Actively quoting ($7M first site → $49M 7-site)

Total signed capacity: 2,165 kg/day across 7 sectors

Total pipeline value: $100M+

6.2 Gross Margin Analysis

Selling Price ($/kg)	LCOH Base Case ($/kg)	Gross Margin ($/kg)	Gross Margin (%)
$15	$2.79	$12.21	81.4%
$20	$2.79	$17.21	86.1%
$25	$2.79	$22.21	88.8%
$30	$2.79	$27.21	90.7%

These margins are enabled by the on-site production model: Tobe eliminates the 75–85% of delivered hydrogen cost attributable to transport, storage, and distribution.

6.3 The On-Site Advantage

Source: MKRT_CASE_CostAnalysis_OnSiteAdvantage.pdf

The on-site advantage is quantified by an actual market quote:

- Airgas cylinder delivery quote (September 2024): Size 300 high-pressure cylinder, 2,500 psi fill, ~0.693 kg H_2 per cylinder, $84.05 per fill = $121.31/kg delivered
- Tobe on-site production cost: $2.79/kg (base case LCOH)
- Cost reduction vs. cylinder delivery: >97%

Even against lower-cost tube trailer delivery ($15–25/kg for grey hydrogen), Tobe's economics are compelling:

Delivery Method	Customer Pays	Tobe Offers	Customer Savings
Cylinder (Airgas)	$121/kg	$20/kg	83%
Tube trailer (grey H_2)	$15–25/kg	$15–25/kg	0% cost, gains green H_2
On-site PEM (competitor)	$4–7/kg production + CAPEX	$15–25/kg all-inclusive	No CAPEX, no operations burden

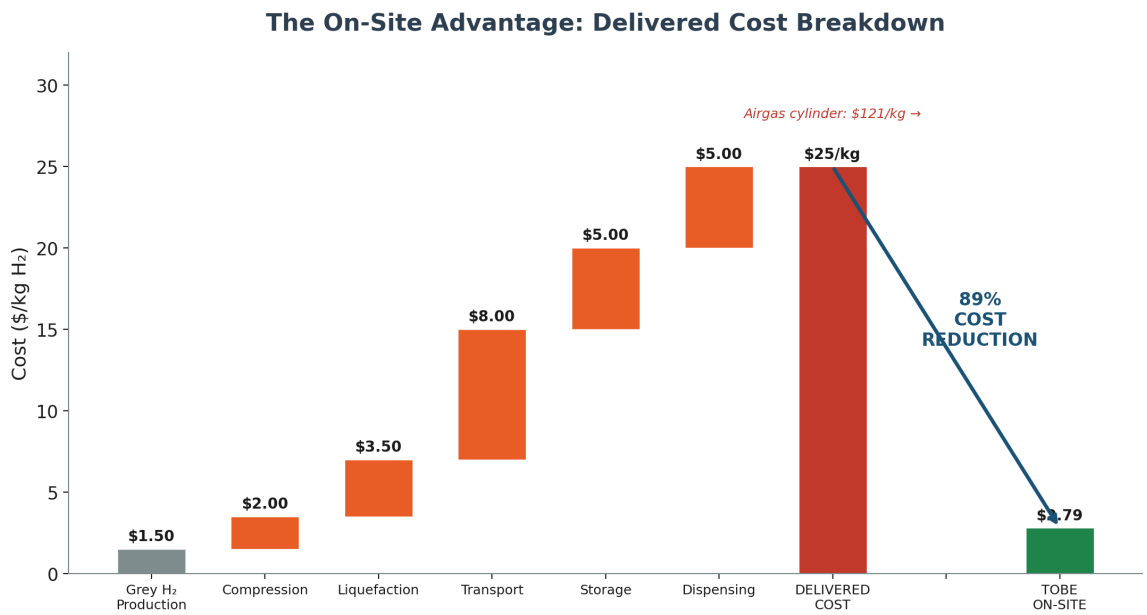

Figure 2: On-Site Production Cost Advantage

The value proposition is that Tobe produces on-site more cheaply than any current delivered hydrogen option while offering green, emission-free hydrogen and eliminating customer CAPEX and operational complexity.

6.4 Delivered Cost by End-Use Application

LCOH at electrolyzer outlet is the primary comparability metric. Actual delivered cost to end-use equipment depends on downstream conditioning requirements:

End-Use Application	Outlet Pressure	Compression Adder	Total Delivered Cost
Industrial low-pressure (burners, furnaces, process heat)	~40 psig (outlet)	$0/kg	$2.79/kg
Industrial with buffer storage (batch processes)	200–500 psig	$0.50–1.00/kg	$3.29–3.79/kg
Mobility at 350 bar (bus/truck fueling)	350 bar	$1.50–2.00/kg	$4.29–4.79/kg
Mobility at 700 bar (passenger vehicle fueling)	700 bar	$2.50–3.50/kg	$5.29–6.29/kg

All currently signed customers are low-pressure industrial users — they consume hydrogen at or near the electrolyzer outlet pressure with no compression required. The January 2026 equipment quote confirms this boundary by listing compression and storage as separate TBD line items.

Compression adder estimates based on DOE compression energy ranges (4.5–7.1 kWh/kg for high-pressure applications) and published station-level TEA data (NREL).

6.5 Revenue Streams

The business model includes two complementary revenue streams:

7. Hydrogen offtake (recurring): Produce and sell H_2 on-site at customer locations
8. System sales (one-time + service): Sell electrolyzer units directly

System sales backlog:

- 10× T-25 units: $1.5–2.5M
- 10× T-125 units: $4.5–6.5M
- Total system sales: $6–9M

Source: TEA_PIPELINE_NARRATIVE.md

7. Financial Model / DCF

7.1 LCOH Calculation Methodology

LCOH is calculated using the Capital Recovery Factor (CRF) method, following the approach recommended by Agora Energiewende and the European Hydrogen Observatory LCOH Calculator:

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LCOH = (CAPEX_ann + OPEX_fix + C_replacement + C_electricity + C_water) /
H₂_annual
```

```
Where:

  CAPEX_ann = Installed_CAPEX × CRF

  CRF = r(1+r)^n / ((1+r)^n - 1)

  H₂_annual = Rated_output × 365 × Capacity_factor
```

7.2 Base Case LCOH Calculation (Single T-25 Unit)

Parameter	Value	Source
Rated power	50 kW	T-25 specification
H_2 output	25 kg/day	ENG_HMB
SEC	46.5 kWh/kg (system-level)	Base case (Section 3.3)
Installed CAPEX	$1,000/kW = $50,000	Section 4.2
Electricity price	$0.04/kWh	PSO large-power industrial tariff
Capacity factor	85%	Validated by signed offtake (2,165 kg/day pipeline)
Economic life	20 years	Conservative; no membrane degradation supports longer life
WACC	8%	Base case per deep research report recommendation
Fixed O&M	3% of CAPEX/year	Section 5.2
Stack replacement	10% of CAPEX at 80,000 operating hours	Section 5.4
Water cost	$0.02/kg H_2	Section 5.3

Step-by-step calculation:

Step	Calculation	Result
Annual H_2 production	25 × 365 × 0.85	7,756 kg/year
CRF (8%, 20 years)	0.08 × 4.661 / 3.661	0.1019
Annualized CAPEX	$50,000 × 0.1019	$5,094/year
Fixed O&M	$50,000 × 0.03	$1,500/year
Electricity cost	7,756 × 46.5 × $0.04	$14,426/year
Water cost	7,756 × $0.02	$155/year
Replacement provision	$5,000 / 10.7 years (annualized)	$467/year
Total annual cost		$21,642/year
LCOH	$21,642 / 7,756	$2.79/kg

LCOH Cost Breakdown (Base Case)

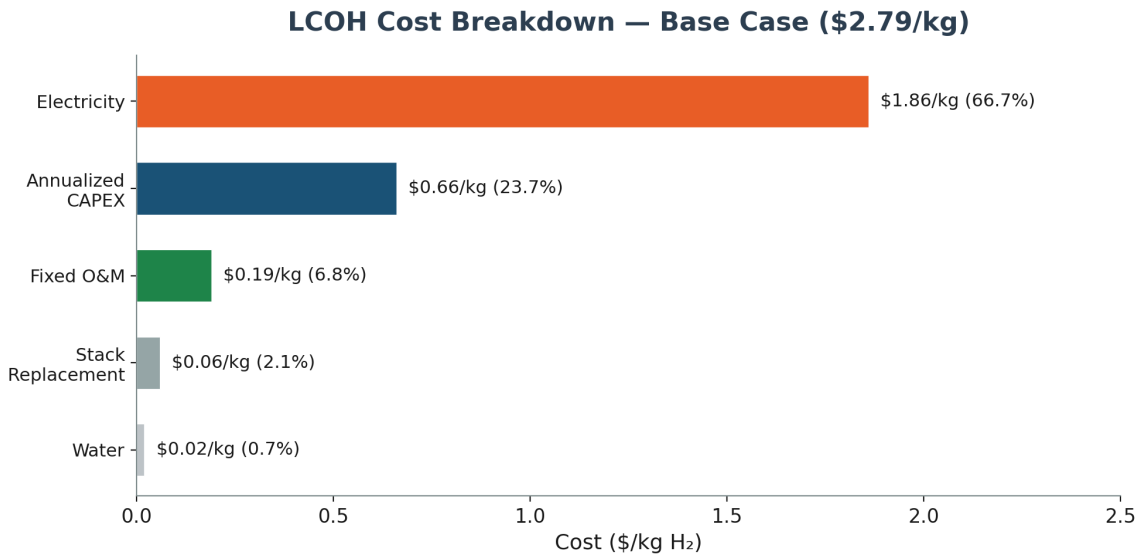

Figure 3: LCOH Cost Breakdown (Base Case)

Electricity dominates at 67% of LCOH, consistent with the 55–70% benchmark from the deep research report. Including the degradation adjustment (+$0.04/kg, Section 5.4), the all-in LCOH is $2.83/kg.

7.3 Three-Scenario LCOH Summary

Parameter	Downside	Base Case	Upside
SEC (kWh/kg)	47.5	46.5	42.0
Electricity price ($/kWh)	0.045	0.040	0.035 (PSO renewable)
Installed CAPEX ($/kW)	1,200	1,000	800
Capacity factor	75%	85%	90%
WACC	10%	8%	6%
Fixed O&M (% CAPEX)	4%	3%	2%
LCOH ($/kg)	$3.61	$2.79	$2.06

All three scenarios remain well below the PEM electrolysis benchmark of $4–7/kg.

7.4 WACC Sensitivity

WACC	CRF (20 yr)	Annualized CAPEX	LCOH
6%	0.0872	$4,360	$2.70/kg
8%	0.1019	$5,094	$2.79/kg
10%	0.1175	$5,873	$2.89/kg

LCOH is relatively insensitive to WACC because CAPEX is only ~24% of total cost. A 4-percentage-point swing in WACC changes LCOH by only $0.19/kg.

7.5 45V Production Tax Credit Analysis

Tobe qualifies for the Section 45V Clean Hydrogen Production Tax Credit. Final rules under 45VH2-GREET determine the credit tier based on lifecycle emissions. Three scenarios are modeled reflecting compliance uncertainty:

45V Credit Scenarios

Scenario	Credit Value	Qualification Requirements	Compliance Cost	Effective Base LCOH
Full — Tier 1	$3.00/kg	Prevailing wage + registered apprenticeship + lifecycle emissions ≤0.45 kg CO_2e/kg H_2 (45VH2-GREET)	$0.10/kg	-$0.11/kg
Partial — Tier 2	$1.00/kg	Lifecycle emissions 0.45–1.5 kg CO_2e/kg H_2; reduced compliance requirements	$0.05/kg	$1.84/kg
No PTC	$0/kg	Baseline (credit not secured or not eligible)	$0	$2.79/kg

Temporal matching requirements: EAC-based claims require annual matching through 2029 and hourly matching starting 2030. Tobe's grid-connected model in Oklahoma requires an explicit electricity attributes strategy (EACs or direct PPA) for Tier 1 qualification.

Compliance costs ($0.05–0.10/kg) include verification, reporting, EAC procurement, and prevailing wage/apprenticeship administration. These are modeled as a direct LCOH adder rather than absorbed into O&M.

The subsidy value per facility under the Full Tier 1 scenario (based on financial model projections):

- $27M per eligible facility over 10-year PTC window
- 4 eligible facilities identified (Tulsa, OKC, Seattle, Spokane)
- Total 45V value: $108M cumulative

Source: Fin_FinancialModel_v4.3.0.xlsx; Tobe_TEA_Narrative_v1.md

Note: 45V PTC qualification requires compliance with prevailing wage and registered apprenticeship requirements under IRA Section 45V(e). The begin-construction deadline is currently 2027 under OBBBA, with a 10-year production credit window. The negative effective LCOH under the Full Tier 1 scenario reflects the PTC's intended policy outcome — incentivizing domestic clean hydrogen production — and should be interpreted as a margin enhancement rather than a sustainable long-term production cost.

7.6 Zeeco Phase 1 Deployment Economics (Worked Example)

The Zeeco ARC deployment provides a concrete validation of the LCOH model against a real contracted project with FEED-level engineering.

Source: TOBE FEED Package – Zeeco ARC; TOBE-PFD-001; TEA_PIPELINE_NARRATIVE.md

Parameter	Value	Source
Configuration	12× T-25 units	TOBE FEED Package
Total capacity	300 kg H_2/day	TOBE-PFD-001
Connected load	576 kW (stacks) + 50 kW (aux) = 626 kW	TOBE FEED Package
Annual H_2 at 85% CF	93,075 kg	300 × 365 × 0.85
Annual electricity	4,327,988 kWh	93,075 × 46.5
Electricity cost	$173,120/year	At $0.04/kWh
Installed CAPEX (est.)	$626,000	626 kW × $1,000/kW
Annualized CAPEX	$63,789	CRF = 0.1019
Fixed O&M	$18,780/year	3% of CAPEX
Water	$1,862/year	93,075 × $0.02
Replacement provision	$5,830/year	10% of CAPEX annualized
Total annual cost	$263,381	
LCOH	$2.83/kg	Slightly above single-unit due to auxiliary load
Revenue at $30/kg	$2,792,250/year	Contracted Zeeco price
Gross profit	$2,528,869/year	
Gross margin	90.6%	
CAPEX payback	~3 months	$626K / $2.53M annual GP

This worked example validates that the unit-level LCOH model scales predictably to multi-unit deployments, and that the contracted Zeeco price of $30/kg produces exceptional margins even at the facility level.

7.7 DCF / Revenue Projections

The financial model projects seven-year revenue growth driven by facility deployment, equipment sales scale-up, and the expanding customer pipeline:

Metric	Year 1 (2026)	Year 2 (2027)	Year 3 (2028)	Year 4 (2029)	Year 5 (2030)	Year 6 (2031)	Year 7 (2032)
H_2 production revenue	$12.4M	$24.1M	$46.1M	$86.6M	$139.6M	$267.2M	$270.0M
Equipment sales revenue	$6.5M	$26.9M	$82.5M	$234.1M	$470.6M	$641.3M	$787.5M
Equipment services revenue	—	$0.4M	$2.3M	$8.5M	$26.6M	$66.7M	$130.1M
45V PTC revenue	$1.5M	$2.9M	$5.3M	$5.4M	$5.4M	$5.4M	$5.4M

Total revenue	$20.4M	$54.3M	$136.2M	$334.6M	$642.2M	$980.6M	$1,193.0M
Gross margin	77.7%	70.4%	68.6%	68.0%	68.7%	71.0%	69.6%
EBITDA	$12.9M	$32.2M	$80.9M	$201.0M	$394.2M	$631.2M	$755.7M
EBITDA margin	63.3%	59.3%	59.4%	60.1%	61.4%	64.4%	63.3%
Net income	$8.7M	$21.7M	$56.3M	$141.5M	$280.6M	$447.1M	$535.1M
Headcount (FTEs)	18	28	37	53	102	120	138
Operating expense	$2.9M	$6.0M	$12.6M	$26.4M	$46.8M	$65.1M	$75.1M
Year-end cash	$12.7M	$77.1M	$149.9M	$330.4M	$603.6M	$1,240.0M	$1,738.8M

Revenue mix shifts from H_2-production-dominant (61% in Year 1) to equipment-sales-dominant (66% by Year 7), reflecting the transition from operated hydrogen-as-a-service deployments to a scaled equipment manufacturing business. Equipment services revenue (recurring maintenance contracts at 75% attach rate) emerges as a growing third stream from Year 2 onward.

Fundraising trajectory:

Round	Timing	Amount	Purpose
Seed	May 2026	$10M ($7.5M equity + $2.5M debt)	First deployments, FEED execution
Series A	June 2027	$40–46M	Multi-site buildout
Series B	March 2029	$60–85M	National scale-up
Series C	March 2031	$150–225M	Full commercialization

Exit valuation framework: $2.6–3.3B at 8–10× revenue multiple

8. Sensitivity Analysis

8.1 Tornado Chart — Single-Variable LCOH Sensitivity

Each variable is varied independently while holding all others at base case ($2.79/kg at $0.04/kWh).

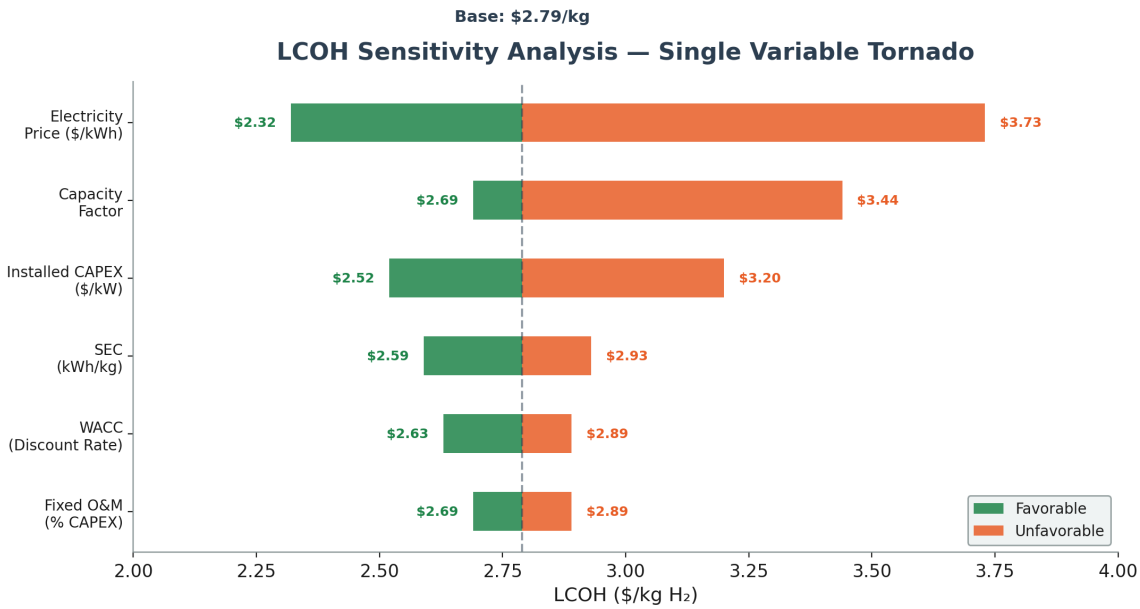

Figure 4: Single-Variable LCOH Sensitivity (Tornado Chart)

Key finding: Electricity price is the dominant sensitivity driver, consistent with the report's finding across all nine exemplar TEAs. A swing from $0.025 to $0.055/kWh changes LCOH by $1.40/kg. Even at the most expensive electricity modeled ($0.055/kWh), Tobe's LCOH ($3.49/kg) remains below the PEM benchmark floor of $4/kg.

8.2 Multi-Variable Scenario Analysis

Scenario	Elec ($/kWh)	CF	CAPEX ($/kW)	SEC (kWh/kg)	WACC	LCOH ($/kg)	vs. PEM Benchmark
Best case	0.025	95%	700	42.0	6%	$1.54	61–78% below
Favorable	0.035	90%	800	42.0	6%	$2.06	48–71% below
Base case	0.040	85%	1,000	46.5	8%	$2.79	30–60% below
Conservative	0.045	75%	1,200	47.5	10%	$3.61	10–48% below
Worst case	0.055	50%	1,500	50.0	10%	$5.61	At PEM parity

Tobe's LCOH exceeds PEM benchmarks only in an extreme worst-case scenario combining the highest electricity price, lowest utilization, highest CAPEX, worst efficiency, and highest cost of capital — a scenario that is unlikely given the signed offtake pipeline and Oklahoma's industrial electricity rates.

8.3 Capacity Factor Sensitivity

The signed pipeline of 2,165 kg/day validates high utilization targets. However, LCOH is moderately sensitive to capacity factor because fixed costs (CAPEX annuity, O&M) are spread over fewer production kilograms at lower utilization:

Capacity Factor	Annual H_2 (kg)	LCOH ($/kg)	Gross Margin at $25/kg
30%	2,738	$4.35	82.6%
50%	4,563	$3.39	86.4%
75%	6,844	$2.90	88.4%
85%	7,756	$2.79	88.8%
95%	8,669	$2.70	89.2%

Even at 30% capacity factor — a level typical of intermittent renewable-only systems — Tobe's LCOH ($4.35/kg) remains within the PEM benchmark range. Gross margins exceed 82% at all utilization levels when selling at $25/kg.

8.4 Hydrogen Selling Price Sensitivity

Selling Price ($/kg)	Gross Margin (Base LCOH)	Annual Gross Profit per T-25 (85% CF)	Payback Period
$15	81.4%	$94,661	<7 months
$20	86.1%	$133,441	<5 months
$25	88.8%	$172,221	<4 months
$30	90.7%	$211,001	<3 months

At all contracted/LOI price points, the T-25 unit pays back its installed capital ($50,000 at base case) in under 12 months.

9. Competitive Comparison

9.1 Production Cost Comparison

Technology	SEC (kWh/kg)	Installed CAPEX ($/kW)	LCOH at Outlet ($/kg)	Source
PEM electrolysis	50–58	1,400–2,500	$4.40–7.90	DOE 2024 program record
Alkaline electrolysis	51–56	500–1,000	$4.00–6.00	IRENA 2020; IEA 2023
Grey hydrogen (SMR)	N/A (NG feedstock)	—	$1.00–2.00	DOE reference
Tobe Energy (base)	46.5	1,000	$2.79	This analysis
Tobe Energy (upside)	42.0	800	$2.06	This analysis

9.2 Wall-to-Plug Efficiency Comparison

This comparison requires careful framing. Most published PEM efficiencies are stack efficiencies (DC → H_2), not wall-to-plug (AC → H_2). Tobe reports wall-to-plug exclusively. See Section 3.2.1 for the stack vs. system SEC boundary clarification.

Metric	Tobe Energy (System)	Tobe Energy (Stack)	PEM (Wall-to-Plug)	PEM (Stack Only)
SEC (kWh/kg)	46–48	42.2	50–58	42–51 (DC basis)
LHV efficiency	70–72%	79%	57–67%	65–80%
HHV efficiency	83–85%	93.4%	67–79%	77–95%

Source: ENG_EfficiencyCalculations.xlsx (Tobe); TOBE-PFD-001 (stack SEC); DOE 2024, NREL 2024, IRENA 2020 (PEM)

Tobe's system-level wall-to-plug efficiency of 70–72% LHV exceeds the typical PEM wall-to-plug range of 57–67% LHV. Tobe's stack-level SEC of 42.2 kWh/kg is comparable to the best PEM stack-only numbers — but Tobe achieves this with stainless steel at a fraction of the cost.

9.3 Delivered Cost Comparison (The Comparison That Matters)

For end customers, the relevant comparison is total cost of hydrogen at point of use, not production cost:

Source	Production Cost	Delivery Cost	Total Delivered
Grey H_2 (SMR) + tube trailer	$1–2/kg	$13–23/kg	$15–25/kg
Grey H_2 (SMR) + cylinder (Airgas)	$1–2/kg	$119–120/kg	$121/kg
PEM (on-site, self-owned)	$4–7/kg	$0	$4–7/kg + CAPEX risk
PEM (centralized) + delivery	$4–7/kg	$5–10/kg	$9–17/kg
Tobe (on-site, as-a-service)	$2.79/kg	$0	$15–30/kg all-in

Tobe's value proposition: customers pay $15–30/kg for on-site green hydrogen, which is at or below their current delivered grey hydrogen cost, with zero CAPEX commitment, zero operational burden, and the environmental benefit of emission-free production.

9.4 Structural Advantages Summary

Advantage	Impact	Durability
No membrane	Lower CAPEX, no degradation-driven replacement	Permanent — fundamental architecture
No rare earth catalysts	Lower CAPEX, no supply chain risk (Ir/Pt)	Permanent — stainless steel construction
Near-ambient operation (27°C)	No cooling BOP, lower energy consumption	Permanent — thermodynamic characteristic
TDS tolerance (25–10,000+ ppm)	Standard RO vs. UPW/DI, broader site applicability	Permanent — demonstrated across conditions

On-site production model	Eliminates 75–85% of delivered H_2 cost for low-pressure uses	Structural — business model advantage
Wall-to-plug efficiency leadership	Lower electricity cost per kg at the dominant cost driver	Improvable — optimized topology shows 94% HHV path

10. Conclusions

10.1 Tobe's measured efficiency already beats PEM benchmarks

Across four test configurations spanning two power levels and two water quality conditions, Tobe's electrolyzer delivers 70–72% LHV wall-to-plug efficiency (system SEC of 46–48 kWh/kg; stack SEC of 42.2 kWh/kg). This exceeds the typical PEM wall-to-plug range of 57–67% LHV, using conservative bench power supply conditions. The demonstrated upside (94% HHV with resonant LLC topology) suggests significant headroom for further improvement.

Source: ENG_EfficiencyCalculations.xlsx; ENG_PreCommit_SampleOperatingData.xlsx; TOBE-PFD-001

10.2 On-site production eliminates 75–85% of delivered hydrogen cost

DOE data and the Airgas market quote ($121/kg delivered) confirm that transport, storage, and distribution dominate hydrogen cost for end users. Tobe's on-site model eliminates these costs entirely for low-pressure industrial applications, allowing production-cost LCOH of $2.79/kg to translate directly into customer value at $15–30/kg selling prices.

Source: MKRT_CASE_CostAnalysis_OnSiteAdvantage.pdf; TEA_PIPELINE_NARRATIVE.md

10.3 Real contracted pricing validates >81% gross margins

Gross margins of 81–91% are calculated from real contracted prices ($30/kg Zeeco) and signed LOIs ($15–25/kg across four customers) against a measured production cost basis. The weighted average selling price across signed commitments is ~$19/kg.

Source: TEA_PIPELINE_NARRATIVE.md; Fin_FinancialModel_v4.3.0.xlsx

10.4 Technology advantages drive structural CAPEX advantage

Equipment cost of $500/kW (January 2026 quote) is 37–63% below PEM equipment-only cost of $800–1,350/kW (NREL 2024). Fully installed at $1,000/kW, Tobe remains below the PEM uninstalled cost. This advantage stems from elimination of the membrane, rare earth catalysts, and active cooling system, combined with commodity stainless steel construction.

Source: January 2026 quote (Jan 2026); TOBE-FEED-PKG-001; NREL 2024 (deep research report)

10.5 $100M+ pipeline across 7 sectors proves horizontal market applicability

With 2,165 kg/day of signed capacity across industrial, manufacturing, academic, mobility, e-fuels, transit, and EPC, Tobe has demonstrated that the on-site hydrogen production model addresses a broad, horizontal market — not a single niche application.

Source: TEA_PIPELINE_NARRATIVE.md

10.6 Investment Thesis

Tobe Energy offers investors:

9. Production economics that beat all current electrolysis benchmarks ($2.79/kg vs. $4–7/kg PEM)
10. Gross margins of 81–91% validated by real commercial contracts
11. A $100M+ pipeline across seven industry sectors with geographic diversity
12. Technology differentiation that is structural (no membrane, no rare earths, near-ambient operation) rather than incremental
13. Policy tailwinds via the 45V PTC (up to $108M cumulative value across four facilities under Full Tier 1)
14. Capital efficiency with sub-12-month payback on deployed units

Appendix A: Source File Reference

File	Description	Key Data Used
ENG_PreCommit_SampleOperatingData.xlsx	22,103 data points, 5-sec intervals, 6kW prototype	System SEC (46.08 kWh/kg), stack temp (27–28°C), operating pressure
ENG_EfficiencyCalculations.xlsx	Four test runs across 2 power levels, 2 TDS conditions	LHV efficiency (69–72%), HHV efficiency (82–85%)
ENG_HMB_HeatAndMaterialBalance.xlsx	T-25 unit H&MB + multi-unit scaling	Water consumption, energy inputs, material flows
ENG_GRANT_NavySBIR_N242-070.pdf	Published efficiency data (federal grant)	Efficiency validation, TDS tolerance data
TOBE FEED Package – Zeeco ARC.pdf	Real EPC scope and installed cost data	System boundary, equipment scope, battery limits
TOBE-FEED-PKG-001.pdf	Detailed FEED engineering package	Materials (304 SS), codes/standards, electrical specs
TOBE-PFD-001.pdf	Process flow diagram	Stack SEC 42.2 kWh/kg deployment basis
Quote (Jan 2026)	Equipment pricing at breakeven	T-25 $25K ($500/kW), T-125 $120K ($480/kW), scope definition
MKRT_CASE_CostAnalysis_OnSiteAdvantage.pdf	On-site vs. delivered cost analysis	Airgas quote ($121/kg), market segmentation
Fin_FinancialModel_v4.3.0.xlsx	Latest financial model (v4.3.0)	Revenue projections, unit pricing, 45V analysis
FINAL_Tobe_Seed_Financial_Model_v853.xlsx	Seed financial model (v8.5.3)	Facility CAPEX, headcount, OpEx trajectory
TEA_PIPELINE_NARRATIVE.md	Customer pipeline, pricing, LOIs	Contracted pricing, utilization validation
TEA_Deep_Research_Report.md	9 TEA exemplars, methodology guide	LCOH methodology, benchmark parameters, sensitivity approach

Tobe_TEA_Narrative_v1.md	Prior TEA narrative	Unit economics, fundraising trajectory
NOTES_operating_data.md	Context for operating data	Critical test parameters
NOTES_efficiency_calculations.md	Context for efficiency data	SBIR label correction, scenario definitions
NOTES_heat_material_balance.md	Context for H&MB data	Zeeco 12-unit revision, deployment timeline

Appendix B: Glossary

Term	Definition
BOP	Balance of Plant — auxiliary systems (power electronics, water treatment, thermal management, controls)
CF	Capacity Factor — fraction of rated output achieved annually
CRF	Capital Recovery Factor — annualization factor for CAPEX
FLH	Full Load Hours — equivalent hours at rated capacity per year
HHV	Higher Heating Value of hydrogen (39.41 kWh/kg = 142 MJ/kg)
LCOH	Levelized Cost of Hydrogen — annualized total cost per kg produced
LHV	Lower Heating Value of hydrogen (33.33 kWh/kg = 120 MJ/kg)
LLC	Inductor-Inductor-Capacitor — resonant power converter topology
PEM	Proton Exchange Membrane — conventional electrolysis technology
PTC	Production Tax Credit — Section 45V of the Inflation Reduction Act
RO	Reverse Osmosis — water purification technology
SEC	Specific Energy Consumption — kWh per kg H_2 produced
SMR	Steam Methane Reforming — conventional hydrogen production from natural gas
TDS	Total Dissolved Solids — measure of water impurity (ppm)
WACC	Weighted Average Cost of Capital — discount rate for DCF analysis